Risk Management

Equity Option and Fixed Income Portfolio Risk

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Asset-Normal VaR(99%, 1-day): Option Portfolio Exposure

Applied the Delta-Normal method assuming joint normality of returns.

The monetary delta for each option:
$x_i = \Delta_i \times S_i \times \text{Contracts}_i \times 1000$.

While efficient, this method may underestimate risk due to its simplifying assumptions and lack of option non-linearity.

Metric		Value (CHF)
Portfolio σ	$\sigma_p = \sqrt{(x^t \cdot \Sigma \cdot x)}$	3'657'708
VaR(99%, 1-day)	$\text{VaR} = 2.33 \times \sigma_p$	8'509'102
ES(99%, 1-day)	$\text{ES} = 2.665 \times \sigma_p$	9'748'576

SMI-Mapped VaR(99%, 1-day): Market Risk via Beta Exposure

Mapped the equity option exposure to the Swiss Market Index (SMI) using asset betas to estimate systemic risk. The adjusted exposure was computed as:
$\text{Exposure}_i = \Delta_i \times S_i \times \text{Contracts}_i \times \text{Beta}_i \times 1000$.

Used the estimated daily volatility of SMI:
$\sigma_{smi} \approx 1.2\%$ and calculated Portfolio standard deviation:
$\sigma_p = |x_{smi}| \times \sigma_{smi}$

This method links portfolio risk to overall market movements but fails to capture specific or nonlinear components, resulting in lower risk estimates.

Metrics		**Value (CHF)**
SMI-adjusted Exposure	**$x_{smi} = \Sigma(\text{Exposure}_i)$**	1'349'877
VaR(99%, 1-day)	**$\text{VaR} = 2.33 \times \sigma_p$**	3'140'283
ES(99%, 1-day)	**$\text{ES} = 2.665 \times \sigma_p$**	3'597'71

Historical Simulation VaR(99%, 1-day)

Simulated 250 daily return scenarios using historical data for NESN and SREN, repricing the portfolio under each shock.

This approach effectively captures fat tails and nonlinear exposure, providing a realistic picture of potential losses.

The high VaR and ES values highlight the method's strength in identifying extreme downside risk beyond parametric models.

1st percentile Return	-51,499%
VaR(99%, 1-day)	14'536'820,70 CHF
Mean Below Var:	-63,110%
ES(99%, 1-day)	17'813'658,16 CHF

Pricing the Fixed Income Portfolio

Priced the fixed income portfolio by discounting each bond's cash flows using the latest ZCB yield curve as of April 25, 2025.

Each bond was valued as the sum of discounted future coupons and face value, assuming semi-annual payments.

The total market value of the USD-denominated bond portfolio amounts to $429.86 million.

# Bonds	Maturity	Coupon	Price (CHF) Σ (Coupon / $(1 + r/n)^t$) + Face / $(1 + r/n)^n$	Total Value (USD)
2,000,000	1 year	0 CHF	96.20	192'400'192
-1'500'000	3 years	5 CHF	103.98	-155'971'425
4'000'000	5 years	2 CHF	98.36	393'430'620

Portfolio value π :	$429’859’387

Asset-Normal VaR(99%, 1-day): Fixed Income Portfolio

Assessed fixed income risk by mapping cash flows to interest rate shocks across 1Y, 3Y, and 5Y maturities.

Using a 250-day yield variance-covariance matrix, i estimated portfolio volatility under the assumption of normally distributed rates, where x is the value exposure to each yield bucket.

This approach is efficient but tends to underestimate losses in the presence of large interest rate movements.

Metrics		Value (CHF)
Portfolio σ	$\sigma_p = \sqrt{(x^t \cdot \Sigma \cdot x)}$	978'748
VaR(99%, 1-day)	$VaR = 2.33 \times \sigma_p$	2'276'909
ES(99%, 1-day)	$ES = 2.665 \times \sigma_p$	2'608'574

Historical Simulation VaR(99%, 1-Day): Fixed Income Perspective

Simulated 250 historical yield shifts (1Y, 3Y, 5Y) to reprice the bond portfolio and capture interest rate shocks.

Resulting VaR and especially ES ($5.45M) are higher than under parametric assumptions, revealing greater tail risk.

Confirms historical simulation better reflects extreme scenarios and nonlinear dynamics in fixed income markets.

1st percentile return	-0.5987%
VaR(99%,1)	$ 2'416'697
Mean below VaR	-1.35%
ES(99%, 1-day)	$ 5'446'896

TOTAL FUND RISK

Aggregated the risk of both portfolios assuming zero correlation and converted all figures to CHF.
Results are stable across both scenarios, with only a slight increase in risk under historical simulation.
This confirms the robustness of the estimates and the limited impact of currency conversion.

Metrics	Value ($)
1st percentile Return	-3.49%
VaR(99%,1)	15’090’501
Mean Below VaR	-5.87%
ES(99%,1)	25’368’989

Metrics	Value (CHF)
1st percentile Return	-4.39%
VaR(99%,1)	16’835’500
Mean Below VaR	-5.94%
ES(99%,1)	22’810’995

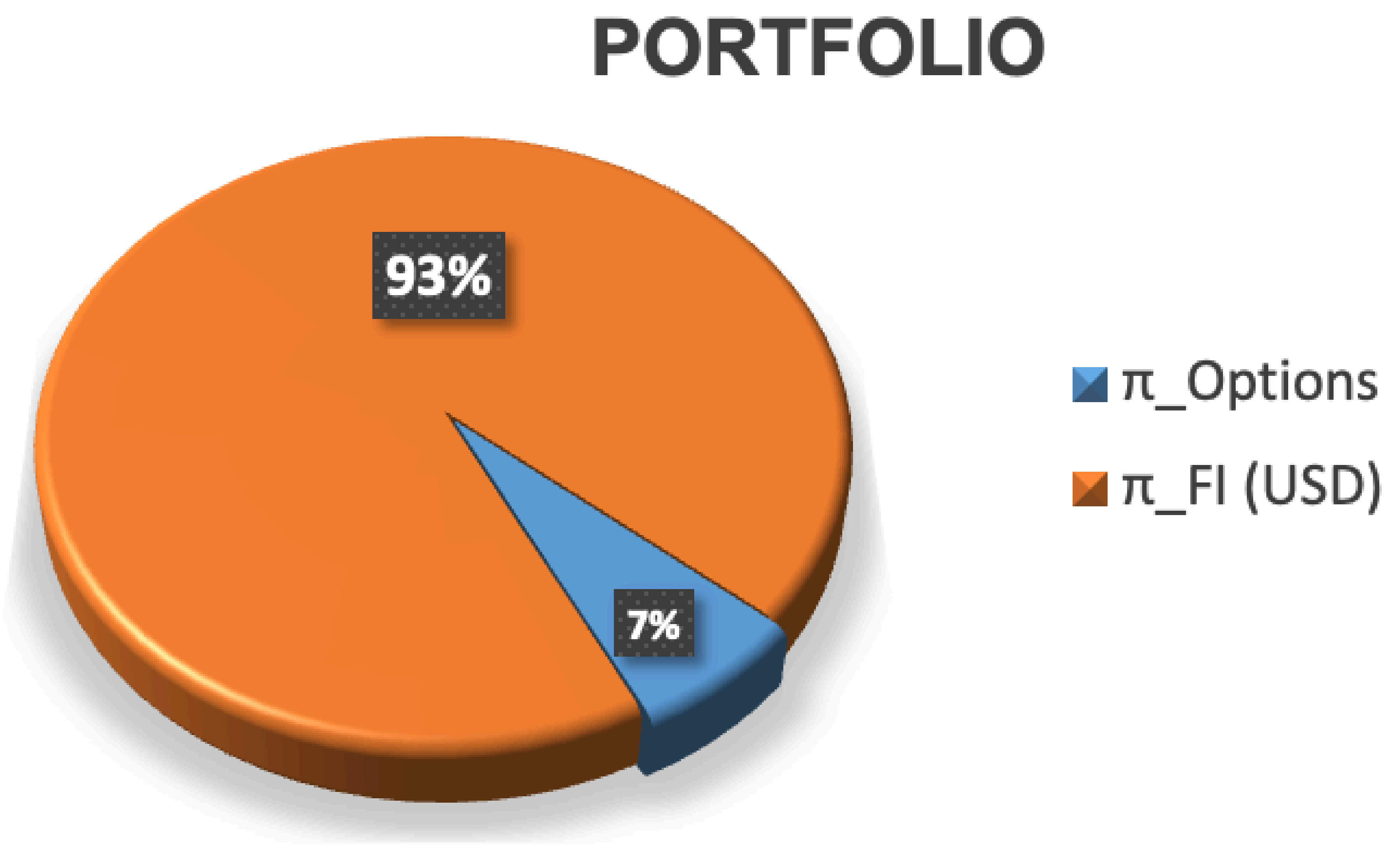
TOTAL FUND RISK

Visual Representation

The portfolio is largely fixed income (93%), with limited exposure to options (7%).

By applying different risk models, i assessed both individual and aggregated portfolio risks.

Results confirm the relevance of model selection and show that historical methods better capture extreme events, with minimal impact from currency conversion.



Thank you for your attention!

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